

Notes: Coase (Economica, 1937)

The Nature of the Firm

Contents

1	Big picture	2
2	Core concepts and measurement analogues	2
2.1	Market coordination versus firm coordination	2
2.2	Costs of using the price mechanism	2
2.3	The employment contract	3
2.4	The size of the firm	3
2.5	Why firm expansion stops	3
2.6	Combination and integration	3
3	Models and analytical specifications	4
3.1	Coase's market-versus-firm comparison	4
3.2	Cost of using the market	4
3.3	Cost of organizing inside the firm	4
3.4	Marginal boundary condition	4
3.5	Long-term contract logic	4
3.6	Firm size comparative statics	5
3.7	Product-scope condition	5
4	Theory design and credibility of the argument	5
4.1	What identifies the mechanism	5
4.2	Why uncertainty alone is not enough	5
4.3	Why the division of labor is not enough	5
4.4	Why cost curves alone are not enough	6
4.5	Why the legal discussion matters	6
4.6	What the paper cannot fully solve	6
5	Tables and figures	6
5.1	Original figures and tables in the paper	6
5.2	Diagram: von Thünen rings and the multi-product boundary of the firm	7
6	Short summary	7
7	Conclusion	8
7.1	Core conclusion	8
7.2	Economic intuition	8
7.3	Takeaway	8

1 Big picture

- **Question.** Why do firms exist in a market economy if the price mechanism can coordinate production?
- **Core puzzle.** Economic theory often treats the market as an automatic price-coordinated system, yet inside a firm a worker moves because someone tells him to move, not because of a relative price change.
- **Main answer.** Firms arise because using the market is costly. The price mechanism requires discovering prices, negotiating and concluding separate contracts, and repeatedly adapting contracts as circumstances change.
- **Definition of a firm.** A firm is the system of relationships that comes into existence when the direction of resources is dependent on an entrepreneur rather than directly on prices.
- **Contractual essence.** Employment replaces many market contracts with one broader contract: the factor agrees, for remuneration, to obey the entrepreneur within specified limits.
- **Boundary of the firm.** The entrepreneur expands the firm until the cost of organizing an extra transaction inside the firm equals the cost of using the market or the cost of organizing that transaction in another firm.
- **Why not one giant firm?** Internal organization also has costs. As the firm grows, managerial costs rise, mistakes become more likely, and factor prices may rise because small firms offer advantages that large firms cannot.
- **Dynamic implication.** Technology and communication change firm size by changing the relative cost of internal organization versus market exchange. The telephone and telegraph, for example, can increase firm size by lowering the cost of spatially dispersed organization.
- **Theoretical contribution.** The paper bridges two assumptions economists use separately: resource allocation by market prices and resource allocation by entrepreneurial direction.
- **Economic takeaway.** The firm is not a production function alone. It is a governance structure that economizes on transaction costs when markets are costly to use.

2 Core concepts and measurement analogues

2.1 Market coordination versus firm coordination

- **Market.** Outside the firm, production is coordinated by a sequence of exchange transactions and relative price signals.
- **Firm.** Inside the firm, those market transactions are superseded by an entrepreneur-coordinator who directs resources.
- **Key contrast.** The same allocation problem can be solved either through prices or through direction.

Interpretation. Coase's contribution is to make the boundary between market and firm an object of economic choice. The relevant question is not whether planning exists, but where planning is cheaper than prices.

2.2 Costs of using the price mechanism

- **Price discovery costs.** Market users must discover relevant prices; price information is not free.
- **Contracting costs.** Each market transaction requires negotiating and concluding a separate contract.

- **Repeated contracting costs.** If repeated short-term contracting is necessary, the fixed cost of making each contract accumulates.
- **Incomplete specification.** Long-term contracts cannot specify all future actions because future circumstances are hard to forecast.
- **Regulatory/tax wedges.** Some taxes or controls apply to market transactions but not to the same transactions carried out inside a firm, which can encourage internal organization.

Interpretation. The paper's famous idea is that the market is not free to use. Once the price mechanism has operating costs, an organization that replaces many market transactions with direction can be efficient.

2.3 The employment contract

- A factor of production employed by a firm enters a contract in which it receives remuneration and agrees to obey the entrepreneur's directions within limits.
- The contract states the limits of authority rather than every future action.
- Within those limits, the entrepreneur decides later what the factor should do.

Interpretation. This is the reason employment is central in Coase's theory. It economizes on repeated contracting by replacing many narrow transactions with one broad authority relationship.

2.4 The size of the firm

- A firm becomes larger when additional transactions that could have been market transactions are organized by the entrepreneur.
- A firm becomes smaller when the entrepreneur abandons the organization of such transactions and leaves them to the market or another firm.
- Firm size is therefore measured by the set of transactions organized under entrepreneurial direction.

Interpretation. This definition makes firm size analytically tractable. It turns the firm boundary into a marginal comparison of alternative governance costs.

2.5 Why firm expansion stops

- **Rising internal organization costs.** As more transactions are organized internally, the cost of organizing the marginal transaction may rise.
- **Managerial mistakes.** The entrepreneur may increasingly fail to allocate factors to their highest-value uses.
- **Rising factor supply prices.** Workers or other factors may require compensation for the disadvantages of being in a large firm rather than a small one.

Interpretation. Coase's answer to "why not one firm for the whole economy?" is that internal coordination has limits. The firm economizes on market costs only up to the point where internal organization becomes equally costly.

2.6 Combination and integration

- **Combination.** Transactions previously organized by two or more entrepreneurs become organized by one entrepreneur.

- **Integration.** Combination becomes integration when it internalizes transactions that were previously carried out between entrepreneurs on a market.
- **Expansion paths.** A firm can expand by combination, integration, or both.

Interpretation. Coase gives horizontal and vertical expansion a common language: both are changes in which transactions are organized inside a single authority structure.

3 Models and analytical specifications

3.1 Coase's market-versus-firm comparison

$$\text{Organize transaction } t \text{ inside the firm if } C_{\text{firm}}(t) \leq \min\{C_{\text{market}}(t), C_{\text{other firm}}(t)\}. \quad (1)$$

Interpretation. Coase does not write this equation, but it formalizes his marginal logic. A transaction belongs inside the firm only when entrepreneurial direction is cheaper than the available alternatives.

3.2 Cost of using the market

$$C_{\text{market}}(t) = C_{\text{search}} + C_{\text{negotiate}} + C_{\text{contract}} + C_{\text{adapt}} + C_{\text{policy wedge}}. \quad (2)$$

Interpretation. The market price of a good is not the whole cost of market exchange. The buyer must also spend resources finding prices, negotiating, writing contracts, and adapting to new circumstances.

3.3 Cost of organizing inside the firm

$$C_{\text{firm}}(t; n) = C_{\text{direction}} + C_{\text{monitor/communicate}} + C_{\text{error}} + C_{\text{factor compensation}}, \quad (3)$$

where n is the number or complexity of transactions already organized inside the firm.

Interpretation. Internal organization is not free. As n rises, direction becomes harder, mistakes become more likely, and the firm may have to compensate factors for disadvantages of large-scale organization.

3.4 Marginal boundary condition

$$C_{\text{firm}}(t_{n+1}; n+1) = \min\{C_{\text{market}}(t_{n+1}), C_{\text{other firm}}(t_{n+1})\}. \quad (4)$$

Interpretation. The firm expands until the next internal transaction is just as costly as market exchange or organization by another entrepreneur. This is Coase's marginal theory of firm size.

3.5 Long-term contract logic

$$\text{Many short contracts} \rightarrow \text{high repeated contracting costs}, \quad (5)$$

$$\text{One broad contract} \rightarrow \text{lower contracting costs but requires authority within limits}. \quad (6)$$

Interpretation. The firm appears when future contingencies are difficult to specify but can be handled by granting the entrepreneur limited authority to decide later.

3.6 Firm size comparative statics

Firm size increases when market transaction costs rise or internal organization costs fall, (7)

Firm size decreases when market transaction costs fall or internal organization costs rise. (8)

Interpretation. Technologies that reduce internal communication costs can make firms larger. Technologies that make markets easier to use can make firms smaller.

3.7 Product-scope condition

Add nearby product A or C when organizing it is cheaper than further expansion in current product B . (9)

Interpretation. Firm scope is not determined by a single-product cost curve. A firm may become multi-product because the marginal cost of organizing a different but nearby transaction can be lower than the marginal cost of expanding the original activity.

4 Theory design and credibility of the argument

4.1 What identifies the mechanism

- The paper starts from the observed fact that markets and firms are alternative ways to coordinate production.
- The mechanism is identified by asking what makes entrepreneurs substitute direction for prices.
- The answer is relative cost: firms arise when authority saves the costs of market exchange.

Interpretation. Coase's logic is a comparative institutional analysis. It compares the cost of two real-world mechanisms rather than assuming one of them is costless.

4.2 Why uncertainty alone is not enough

- Knight emphasized uncertainty, residual risk-bearing, and guaranteed wages.
- Coase accepts that uncertainty is relevant but argues that it does not explain why prices are superseded.
- One can sell advice, make futures contracts, or guarantee a result without directing the counterparty's work.

Interpretation. The missing element in the uncertainty story is why authority replaces market contracts. Transaction costs provide that missing link.

4.3 Why the division of labor is not enough

- Some earlier accounts viewed the firm as the integrating force required by the division of labor.
- Coase's objection is that the price mechanism already integrates specialized activities.
- The real question is why one integrating force, entrepreneurial direction, is sometimes substituted for another, the price mechanism.

Interpretation. Specialization explains the need for coordination, but not why coordination is done inside firms rather than through markets.

4.4 Why cost curves alone are not enough

- An upward-sloping one-product cost curve does not determine the size of the firm.
- A firm can produce multiple products, and the marginal internal organization cost may be lower for a different transaction than for more of the same transaction.
- The boundary depends on marketing costs and organization costs across heterogeneous transactions.

Interpretation. Firm size is a governance problem, not only a production technology problem.

4.5 Why the legal discussion matters

- Coase compares his economic definition with the legal relationship of master and servant.
- The legal distinction turns on the right to control the worker's activity within the terms of service.
- This matches the economic concept of a firm as direction within contractual limits.

Interpretation. The theory is realistic because it maps closely onto legal and practical distinctions between employees and independent contractors.

4.6 What the paper cannot fully solve

- The paper gives a general theory of firm boundaries but does not quantify transaction costs.
- It does not provide a formal empirical test or a dataset.
- It treats internal organization costs at a high level rather than modelling incentive conflicts in detail.
- It does not develop a complete theory of ownership, residual claims, or agency costs.

Interpretation. The paper is foundational because it reframes the problem. Later theories of transaction costs, property rights, and agency build on the questions Coase opened.

5 Tables and figures

5.1 Original figures and tables in the paper

- The article has no empirical tables.
- The paper contains one original diagram, reproduced directly below from the paper.

5.2 Diagram: von Thünen rings and the multi-product boundary of the firm

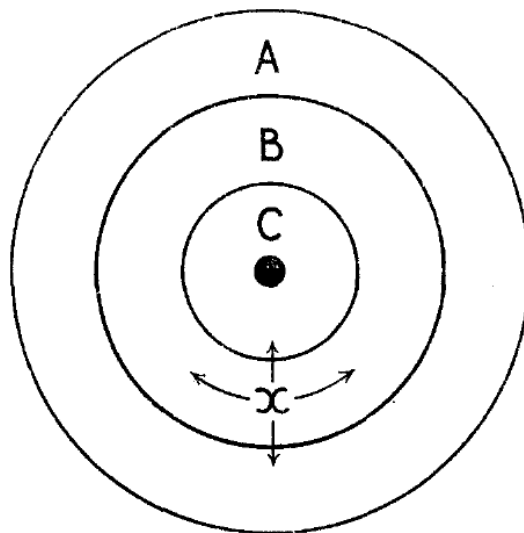


Figure 1: Original diagram directly cropped from Coase (1937), p. 402.

Read:

- The diagram shows a consuming center surrounded by industries *A*, *B*, and *C* in rings.
- The entrepreneur starts from point *x*.
- As the entrepreneur expands in product *B*, the internal cost of organizing more *B* transactions rises.
- At some point, it may be cheaper to organize a nearer but different product, such as *A* or *C*, than to organize more of *B*.

Economic message. The size and scope of a firm cannot be inferred from the cost curve of a single product. A firm chooses among heterogeneous transactions, products, and locations. The boundary is where internal organization stops being cheaper than the market or another firm.

6 Short summary

- The paper asks why firms exist if the price mechanism can coordinate economic activity.
- Coase's answer is that using prices has costs: price discovery, negotiation, contracting, and adaptation.
- A firm replaces many market transactions with an authority relationship governed by a broad contract.
- The firm is defined by the direction of resources by an entrepreneur within contractual limits.
- The firm expands when internal organization is cheaper than market exchange.
- The firm stops expanding when internal organization costs equal the cost of market exchange or organization by another firm.
- Internal costs rise because managers face diminishing returns, make mistakes, and may have to pay more for factors.
- The theory explains vertical integration, combination, product scope, and changes in firm size.

- Coase rejects explanations based only on uncertainty, the division of labor, or conventional cost curves.
- The legal distinction between employees and independent contractors supports the theory because it centers on the right of control.
- The central message is that firms and markets are alternative governance mechanisms, and the boundary between them is determined by relative transaction and organization costs.

7 Conclusion

7.1 Core conclusion

Coase defines the firm as a system of relationships in which the entrepreneur directs resources, replacing the price mechanism for a set of transactions. Firms exist because the market is costly to use, and they expand only until the marginal internal organization cost equals the cost of market exchange or organization by another firm.

7.2 Economic intuition

Markets are powerful but not free. Every market exchange requires discovering prices, negotiating contracts, and adapting to future uncertainty. A firm saves some of these costs by substituting authority for repeated market bargaining. But authority itself becomes costly as the firm grows. The observed boundary of the firm reflects the trade-off between these two costs.

7.3 Takeaway

The Nature of the Firm turns the firm from a black-box production unit into an institution with an economic purpose. The firm exists because it can sometimes coordinate production more cheaply than the market, but it remains limited because internal coordination also has costs.